

Class #3 - Diagonalization

1. Theorem: For $A \in R^{n \times n}$, if λ is an eigenvalue of A , then
algebraic multiplicity of λ _____ geometric multiplicity of λ .
2. Definition: Let $A, B \in R^{n \times n}$. The matrices A and B are called *similar* if there is an _____ $n \times n$ matrix P such that _____.
Note that in such case, A and B have the same characteristic polynomial.
3. Theorem: _____ matrices have the same _____.

4. Idea: Diagonal matrices are the easiest to deal with. (easy to _____, easy to _____, etc.)
 Question: What kind of matrices are _____ to a _____ matrix?
- Want A such that $A =$ _____ for some P _____.
5. Example: Let A be 5×5 real matrix with three distinct eigenvalues $\lambda_1, \lambda_2, \lambda_3$ with geometric multiplicities 2, 2, 1, respectively.

If geometric multiplicities are 1, 2, 1?

6. Def: An $n \times n$ matrix is called *diagonalizable* if it is similar to a _____ matrix.
7. Theorem: An $n \times n$ matrix is diagonalizable _____ the sum of the _____ multiplicities of _____ the eigenvalues equals to _____.
8. Example: Diagonalize the following matrix, if possible.

$$\begin{bmatrix} 7 & 4 & 16 \\ 2 & 5 & 8 \\ -2 & -2 & -5 \end{bmatrix}$$